

Hybrid Architecture of LLM and HRM for Multi-Step Mathematical Problem Solving.

Interim Progress Review Report
CM 4900 – Research Project in Artificial Intelligence

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2026

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Abstract

Mathematical word problems require a model to read a question, identify the relevant quantities and relationships, and execute multiple reasoning steps to reach a correct answer. Most current models process the problem text directly and do not show how they reached their answer. No existing end-to-end pipeline converts natural language problems into structured reasoning inputs while simultaneously producing interpretable reasoning traces and natural language explanations, this is the gap this research addresses.

The technology extended in this work is the Hierarchical Reasoning Model (HRM), a nested latent reasoning architecture comprising a slow high-level planning module and a fast low-level execution module. HRM provides structured multi-step computation but lacks a natural language front-end and produces no human-readable output.

The proposed solution is a hybrid pipeline that integrates an LLM-based parser, a structured JSON encoder, an HRM reasoning core, and an LLM-based explanation module. The parser converts each GSM8K problem into a structured JSON representation capturing quantities and relations. This structured input seeds the HRM reasoning core, which produces both a final answer and intermediate reasoning traces. The explanation module then converts these traces into a step-by-step natural language explanation.

The hypothesis is that replacing generic text embeddings with structured reasoning inputs, combined with trace-aware reasoning, will improve answer correctness on GSM8K while making the reasoning process interpretable and auditable.

The top-level design consists of five components: LLM-based Parser, Bridge Module (JSON to latent tensors), HRM Core (high-level planning and low-level execution with adaptive halting), Post-Processor (solution trace extraction), and LLM-based Explanation Module (answer and explanation generation).

The HRM reasoning core has been fully implemented and is currently training on the GSM8K benchmark. Evaluation targets a minimum of 40% exact-match accuracy compared to standalone LLM and HRM baselines, with interpretability assessed through explanation coherence.

This work contributes a practical, end-to-end pipeline that replaces raw text-based inputs with structured reasoning features and produces auditable reasoning traces advancing the state of interpretable reasoning in neural models for mathematical problem solving.

1. Introduction

1.1 Introduction

Solving mathematical word problems is one of the most common ways to test whether an AI system can reason step by step. These problems require a model to read a question written in plain English, pick out the important numbers and relationships, and then work through multiple steps to get the right answer. Recent AI models have become quite good at this task, but they often cannot explain how they reached an answer. This makes it hard to trust them or check whether their reasoning was correct . [1][5]

At the same time, large language models have shown that they can parse language and generate structured explanations. However, combining these abilities with a compact, trainable reasoning system in a single pipeline is still an unsolved problem . This research tries to solve that problem by building a pipeline that is both accurate and transparent. [3][6]

1.2 Objectives

The objectives of this research are:

1. To review existing neural reasoning systems and identify their weaknesses in producing structured, interpretable outputs . [2][4]
2. To design a hybrid reasoning pipeline that combines structured language parsing, step-by-step reasoning, and natural language explanation in one system.
3. To implement the pipeline and test it on the GSM8K dataset . [5]
4. To measure whether structured inputs and reasoning traces make the model more accurate and easier to audit.

1.3 Problem in Brief

Most current models take a math word problem as plain text and use a dense vector embedding to guess the answer directly. This approach does not keep track of the individual numbers, variables, or steps involved , it just pattern-matches from input to output. As a result, it is impossible to look inside the model and see why it gave a particular answer. When a problem has several numerical relationships or multiple steps, this kind of model tends to fail or give unreliable answers . The core problem this research addresses is: how can we make a reasoning model both more accurate and more transparent at the same time? [4][5]

1.4 Background and Motivation

Researchers have tried several approaches to improve reasoning in AI chain-of-thought prompting, programme-of-thought, and step-by-step instruction tuning are among the most popular. These methods do improve accuracy, but they usually depend on very large black-box models that give no insight into their internal process. The Hierarchical Reasoning Model (HRM) offers a more structured alternative, it uses a slow planning module and a fast execution module that work together in a loop. However, HRM takes raw text or random latent vectors as input and produces no human-readable explanation of its reasoning. [7]

This gap between high-accuracy reasoning and transparent, auditable outputs is what motivated this project. By adding a structured front-end parser and a back-end explanation module around the HRM core, the pipeline becomes both more interpretable and more aligned with how humans actually solve word problems.

1.5 Novel Approach

This research introduces a hybrid pipeline that connects four components end to end:

- An **LLM-based parser** that reads the word problem and extracts a structured JSON object containing the quantities, variables, and relations in the problem
- A **JSON encoder** that converts this structured representation into vectors the reasoning module can use
- An **HRM reasoning core** that works through the problem in multiple steps, producing a final answer and intermediate reasoning traces
- A **post-processor and explanation module** that turns the reasoning traces into a readable, step-by-step natural language explanation

The hypothesis is that giving the reasoning module structured inputs, instead of raw text embeddings will make it more accurate, and that producing reasoning traces will make the output easier to understand and verify. [3][6]

1.6 Resource Requirements

- **Hardware:** GPU-enabled training environment (NVIDIA T4, 16 GB VRAM - Kaggle Notebooks)
- **Dataset:** GSM8K benchmark - 7,473 training problems, 1,319 test problems (publicly available via HuggingFace)

- **Software:** Python 3.12, PyTorch 2.11, HuggingFace Transformers, datasets library
- **LLM parsing:** API access or locally hosted model for structured JSON extraction from problem text
- **Storage:** JSON cache for pre-parsed problems, model checkpoints across training epochs

1.7 Structure of the Report

This report follows the Milestone Approach structure. Chapter 2 reviews prior work in mathematical reasoning and interpretable AI, identifying the research gap. Chapter 3 examines the core technologies used in the pipeline. Chapter 4 describes the hybrid approach in detail. Chapter 5 covers the system architecture and design decisions. Chapter 6 documents the implementation progress so far.

1.8 Summary

This chapter introduced the problem of opaque reasoning in AI systems for mathematical word problems, explained why it matters, and described how this research proposes to fix it. The key idea is to replace raw text embeddings with a structured representation and to make the reasoning process visible through intermediate traces and natural language explanations. The next chapter reviews existing approaches and shows where this work fits within the wider research landscape.

2. Developments and Challenges in Mathematical Reasoning with AI

2.1 Introduction

This chapter looks at how AI systems for mathematical reasoning have developed over time, what the current best approaches are, what challenges remain unsolved, and where this research fits within that picture. It ends by clearly stating the research gap that this work addresses.

2.2 Early Developments

Early attempts to solve math word problems with computers used rule-based systems and symbolic solvers. These systems worked well on simple problems with fixed templates but broke down as soon as the language varied or the problem structure changed . [4][6]

Recurrent neural networks (RNNs) and sequence-to-sequence models improved things by learning from data rather than hand-written rules. Models like DNS and MathDQN tried to map problem text directly to equations or answer values . [4]

2.3 Current Developments

The introduction of transformer-based large language models changed what was possible. GPT-4, LLaMA, and PaLM can solve many GSM8K problems correctly when given a few example solutions to follow , a technique called chain-of-thought prompting . [7]

Programme-of-thought and PAL-style approaches improve on this by generating executable Python code instead of free text . [5]

The Hierarchical Reasoning Model (HRM) takes a different approach entirely. Instead of generating text, it reasons in a latent vector space using a two-level loop , a slow H-module that plans at a high level and a fast L-module that executes at a low level. This gives HRM a structured internal reasoning process. However, HRM takes raw or randomly initialised vectors as input and produces no output a human can read or check.

2.4 Summary of Research Challenges

The table below summarises the key limitations found across the main approaches:

Approach	What it does well	Key limitation
Rule-based / symbolic	Precise on fixed templates	Cannot handle varied language
RNN sequence-to-sequence	Learns from data	Fails on multi-step problems
Chain-of-thought prompting	Step-by-step text output	Steps are unverified free text
Programme-of-thought / PAL	Executable, checkable code	Requires very large black-box LLM
HRM	Structured internal latent reasoning	No natural language front-end or explanation
Neuro-symbolic systems	Precise and interpretable	Does not scale to open-ended language

Table 2.1: Comparison of existing AI approaches for mathematical reasoning.

2.5 Problem Definition - Research Gap

No existing system combines all three of the following in a single trainable pipeline:

1. A **structured front-end** that converts natural language into an explicit representation of the quantities and relations in the problem
2. A **structured reasoning core** that works through the problem step by step and produces inspectable intermediate states
3. A **back-end explanation module** that converts those intermediate states into a readable, auditable explanation

LLM-based approaches have (1) and (3) but not (2). HRM has (2) but not (1) or (3). No prior work connects all three in one end-to-end trainable pipeline for mathematical word problem solving. This is the gap this research addresses . [1][2][3]

2.6 Technology to Bridge the Gap

The technology extended in this work is the Hierarchical Reasoning Model (HRM) , positioned within the AI body of knowledge under: [1]

Deep Learning → Transformer-Based Architectures → Latent Hierarchical Reasoning

Models

HRM is extended by adding an LLM-based parser at the front , which converts the natural language problem into a structured JSON representation and an LLM-based explanation module at the back , which converts the HRM reasoning traces into a step-by-step natural language explanation. The Bridge Module connects the JSON representation to the HRM latent space.

2.7 Summary

This chapter traced how AI reasoning for mathematical word problems developed from rule-based systems through RNNs to modern LLMs and HRM. It identified that no existing work combines structured parsing, latent hierarchical reasoning, and explanation generation in one pipeline. This gap is precisely what the proposed hybrid architecture is designed to fill. Chapter 3 examines the individual technologies that make up this pipeline in detail.

3. Research Challenges in the Technologies Used for the Hybrid Reasoning Pipeline

3.1 Introduction

This chapter examines the core technologies that make up the proposed hybrid pipeline. For each technology, it explains what it is, where it sits within the AI body of knowledge, what it does well, what its limitations are, and how it is used or extended in this research. Understanding these technologies is important because the research gap identified in Chapter 2 can only be bridged by combining them in a specific way.

3.2 Hierarchical Reasoning Model (HRM)

Position in AI body of knowledge:

Deep Learning → Transformer-Based Architectures → Latent Hierarchical Reasoning Models

Fundamentals:

The Hierarchical Reasoning Model organises reasoning into two levels that run in a loop. The H-module (slow planner) runs once per outer cycle and produces a high-level plan. The L-module (fast executor) runs several steps under that plan, refining the latent state each time. Because the latent state is detached between outer cycles, the model avoids storing the full computation graph in memory. Adaptive Computation Time (ACT) allows the L-module to stop early when it is confident, saving computation on subsequent steps. [1]

Strengths:

HRM provides a genuinely structured internal reasoning process, it does not just pattern-match from input to output. It is memory-efficient and supports variable-depth computation.

Limitations:

HRM takes randomly initialised latent vectors as input and has no way to read or parse natural language directly. It also produces no human-readable output, the final latent state must be decoded externally. [2]

How extended in this work:

This research adds a structured front-end (LLM-based parser and JSON encoder) that feeds the HRM core with meaningful, problem-specific latent vectors instead of random noise. A post-processor and explanation module are added at the back end to convert the HRM reasoning traces into readable natural language. This extends HRM into a complete, interpretable end-to-end pipeline.

3.3 Large Language Models (LLM-based Parser and Explainer)

Position in AI body of knowledge:

Deep Learning → Transformer-Based Architectures → Generative Large Language Models

Fundamentals:

Large language models such as GPT-4 and LLaMA are trained on massive text corpora and can follow instructions, extract structured information, and generate coherent explanations. In this pipeline, an LLM is used in two roles. At the front, it reads each GSM8K problem and extracts a structured JSON object listing the known quantities, variables, and relationships. At the back, it receives the HRM reasoning traces and generates a step-by-step natural language explanation of how the answer was reached. The LLM is accessed via API and its outputs are cached to avoid repeated calls during training. [3]

Strengths:

LLMs understand open-ended natural language well and can produce both structured data and fluent explanations. They generalise across many problem types without task-specific fine-tuning.

Limitations:

They are large, expensive to run, and cannot be trained end-to-end within the pipeline. Their internal reasoning is still opaque the JSON they produce must be validated before use.

How extended in this work:

Rather than using the LLM as the sole reasoning engine, this research uses it only as a parser and explainer , lightweight roles that do not require the LLM to perform arithmetic. The actual reasoning is delegated to the HRM core, which is trainable and inspectable.

3.4 Structured JSON Representation

Position in AI body of knowledge:

Artificial Intelligence → Knowledge Representation → Structured Data Encoding

Fundamentals:

JSON is a lightweight format for representing structured data as key-value pairs. In this research, each GSM8K problem is converted into a JSON object that explicitly lists the quantities mentioned in the problem, their relationships, and any intermediate variables. This structured representation is then encoded into a fixed-size vector by a small encoder network before being passed to the HRM core . [6]

Strengths:

Structured representations make the model's inputs transparent and auditable a human can read the JSON and verify that the parser understood the problem correctly. They reduce ambiguity by separating the numbers from the surrounding language.

Limitations:

Building a reliable parser that always produces valid, complete JSON is non-trivial. Parsing errors at this stage can propagate through the entire pipeline.

How extended in this work:

The JSON encoder converts the structured representation into latent tensors that seed the HRM

reasoning core. This is the Bridge Module in the architecture , the key connection between the language understanding layer and the reasoning layer.

3.5 Adaptive Computation Time (ACT)

Position in AI body of knowledge:

Deep Learning → Dynamic Neural Networks → Adaptive Depth Models

Fundamentals:

Adaptive Computation Time is a mechanism that lets a neural network decide how many processing steps to take for each input, rather than always using a fixed number. A halt gate computes a halt probability at each step. When the cumulative halt probability crosses a threshold, the loop stops early. This means simple problems use fewer steps and complex problems use more. [1]

Strengths:

ACT reduces unnecessary computation on easy problems and allows the model to allocate more processing to harder ones. It makes the number of reasoning steps an observable output, which supports interpretability.

Limitations:

The halt gate adds extra parameters and requires careful tuning of the halt threshold to avoid stopping too early or too late.

How extended in this work:

ACT is used inside the HRM L-module to control how many inner execution steps are taken per outer planning cycle. The halt step count is recorded as a diagnostic output, giving insight into how much computation each problem required.

3.6 PyTorch Training Framework

Position in AI body of knowledge:

Deep Learning → Training Frameworks and Tools

Fundamentals:

PyTorch is an open-source deep learning framework that provides flexible model construction, automatic differentiation, and GPU acceleration. All model components in this project are implemented as PyTorch `nn.Module` classes. The GSM8K dataset is loaded using the HuggingFace datasets library, which handles data splitting and batching. A custom Adam-atan2 optimiser is used, which replaces the standard update rule with an arctangent function to make updates scale-invariant and numerically stable. [5]

Strengths:

PyTorch is widely used in research, has excellent community support, and its dynamic computation graph makes debugging straightforward.

Limitations:

Custom training loops require careful management of memory, gradient flow, and device placement, which adds engineering complexity.

How extended in this work:

The training loop includes auxiliary decoder losses at each HRM H-step and a log1p normalisation scheme for the GSM8K answer range. These are implementation decisions specific to this research, built on top of standard PyTorch components.

3.7 Summary

This chapter examined the five core technologies used in the hybrid pipeline: HRM for structured latent reasoning, LLMs for language parsing and explanation, JSON encoding for structured feature representation, ACT for adaptive computation depth, and PyTorch as the training foundation. Each technology addresses a specific part of the research gap identified in Chapter 2. No single technology alone bridges the gap, it is their combination in a unified pipeline that constitutes the contribution of this research. Chapter 4 describes how these technologies are brought together in the proposed approach.

4. A Novel Approach to Hybrid Architecture for Structured Mathematical Reasoning

4.1 Introduction

This chapter presents the proposed solution to the research gap identified in Chapter 2. It explains what the system does, why it is expected to work, what goes in, what comes out, how it works step by step, who will use it, and what makes it different from existing approaches. The proposed solution is a hybrid pipeline that connects a language model parser, a structured encoder, a hierarchical reasoning core, and an explanation module into one end-to-end system.

4.2 Proposed Extension - Name and Positioning

The proposed system is called the **Hybrid Architecture for Structured Mathematical Reasoning**. It extends the Hierarchical Reasoning Model (HRM) by adding a structured natural language front-end and a human-readable explanation back-end. Within the AI body of knowledge, this contribution sits at the intersection of: [1]

- Generative Large Language Models (parsing and explanation)
- Latent Hierarchical Reasoning Models (structured multi-step computation)
- Knowledge Representation (structured JSON encoding)

No prior work connects all three of these in a single trainable pipeline for mathematical word problem solving. This is the novel contribution.

4.3 Hypothesis

Theoretical foundation:

HRM provides structured internal reasoning but cannot read natural language or explain its outputs. Large language models can read natural language and generate explanations but do not reason in a structured, inspectable way. Combining them, using the LLM only for what it is

good at (parsing and explaining) and the HRM for what it is good at (structured multi-step reasoning) should produce a system that is better than either alone.

Hypothesis:

If each GSM8K problem is first converted into a structured JSON representation by an LLM parser, and this structured input is used to seed an HRM reasoning core, then the system will achieve higher answer correctness than a baseline that uses raw text embeddings, and will produce reasoning traces that a human can read and verify. [3][6]

4.4 Inputs

The system takes two inputs for each problem:

- A natural language mathematical word problem from the GSM8K dataset for example, "Janet has 3 apples and buys 5 more. She gives 2 to her friend. How many does she have?"
- A structured JSON object produced by the LLM parser from that same problem for example:

```
{  
  "known": {"apples_start": 3, "apples_bought": 5, "apples_given": 2},  
  "unknown": "apples_final",  
  "steps": ["add start and bought", "subtract given"],  
  "answer": 6  
}
```

The JSON explicitly lists the quantities, the relationships between them, and the expected steps. This is what replaces the raw text embedding used in conventional pipelines.

4.5 Outputs

The system produces three outputs:

- A **numeric answer** - the predicted final answer to the problem, rounded to the nearest integer
- A **reasoning trace** - the sequence of intermediate states produced by the HRM reasoning core, showing what the model computed at each step
- An **optional natural language explanation** - a step-by-step written explanation generated by the LLM explanation module from the reasoning trace, for example: "First, Janet adds the 3 apples she started with to the 5 she bought, giving 8. Then she gives away 2, leaving 6."

4.6 Process (Actions)

The pipeline runs in five stages in sequence:

Stage 1 - Parsing:

The LLM parser reads the natural language problem and produces a structured JSON object. This is done once per problem and the result is cached so the LLM does not need to be called again during training. The cache makes training fast and reproducible.

Stage 2 - Encoding:

The JSON encoder reads the structured fields `known values`, `variable names`, and `step descriptions` and converts them into a fixed-size vector. This vector is the input to the reasoning module. Unlike a raw text embedding, this vector explicitly represents the numerical quantities and their relationships.

Stage 3 - Reasoning:

The HRM core takes the encoded vector and works through the problem in a two-level loop. The H-module (slow planner) runs once per outer cycle and decides what the next high-level step should be. The L-module (fast executor) runs several steps under that plan, refining the internal

state. Adaptive Computation Time (ACT) allows the inner loop to stop early on simple problems. This continues for N outer cycles, producing a final latent state.

Stage 4 - Decoding:

The decoder reads the final latent state and produces a numeric answer prediction along with a confidence score. The answer is converted back to the original scale using a denormalisation step.

Stage 5 - Post-processing and Explanation:

The reasoning trace the sequence of latent states across all HRM steps is extracted. If an explanation is requested, the LLM explanation module reads this trace and generates a step-by-step natural language explanation of how the answer was reached.

4.7 Users

The system is intended for three groups of users:

- **AI researchers** who are studying hybrid reasoning architectures and need a reproducible, inspectable baseline for mathematical reasoning
- **Developers** building AI-assisted tools for education or decision support who need systems whose outputs can be audited and explained
- **Educators and students** who want to see not just the answer to a problem but a clear, step-by-step explanation of how it was solved

4.8 Features

The key features that make this approach different from existing work are:

- **Structured inputs:** The model receives explicit quantities and relationships, not a compressed text vector. This reduces ambiguity and makes inputs auditable before they enter the reasoning module.
- **Separated reasoning levels:** The HRM core separates high-level planning from low-level execution, making internal reasoning more structured than a single-pass transformer.
- **Traceable outputs:** The reasoning trace is a direct record of what the model computed at

each step it is not generated after the fact but produced as part of the forward pass.

- **Cached LLM parsing:** The LLM is only called once per problem and the result is stored. This keeps training cost low while still providing high-quality structured inputs.
- **Explanation on demand:** The explanation module is optional it can be switched off during training and switched on during evaluation or deployment.

4.9 Technologies

The approach uses the following technologies:

- **HRM core** - hierarchical latent reasoning with H-module, L-module, and ACT halt gate
- **Large language model** - accessed via NVIDIA NIM (OpenAI-compatible API) for structured parsing and explanation generation
- **JSON encoder** - a small feedforward network that converts structured fields into latent vectors
- **PyTorch 2.11** - model construction, training, and GPU acceleration
- **HuggingFace datasets** - GSM8K dataset loading and preprocessing
- **Adam-atan2 optimiser** - scale-invariant parameter updates using the arctangent update rule
- **Kaggle T4 GPU** - training hardware (16 GB VRAM)

4.10 Summary

This chapter described the complete proposed approach. The system takes a natural language problem, converts it into a structured JSON representation using an LLM parser, encodes it into vectors, runs it through an HRM reasoning core, decodes a numeric answer, and optionally generates a natural language explanation from the reasoning trace. The key idea is that making the inputs structured and the reasoning traceable improves both accuracy and interpretability, which is exactly the gap identified in Chapter 2. Chapter 5 translates this approach into a concrete system architecture and design.

5. Analysis and Design

5.1 Introduction

This chapter describes the system architecture that puts the proposed approach into a concrete design. It explains what each component does, why it was designed that way, and how the components connect to each other. This chapter focuses on design decisions and component roles, not on code or algorithms, which are covered in Chapter 6.

The architecture is shown in Figure 5.1. It consists of six components arranged as a top-to-bottom pipeline: the LLM-based Parser Module, the Bridge Module, the Hierarchical Reasoning Model core, the Post-Processor Module, the LLM-based Explanation Module, and the Final Output.

5.2 Top-Level Architecture

The system follows a single, linear flow from input to output. Each component receives the output of the one above it, processes it, and passes the result downward. There are no loops between components the pipeline runs once per problem from top to bottom.

The flow is:

User Input (natural language problem) → LLM-based Parser Module → Bridge Module → HRM Core → Post-Processor Module → LLM-based Explanation Module → Final Output (answer and explanation)

This linear design was chosen deliberately. Each component has a single, well-defined job. This makes it easy to test each part on its own, replace one component without changing the others, and trace exactly where an error occurred if the system gives a wrong answer.

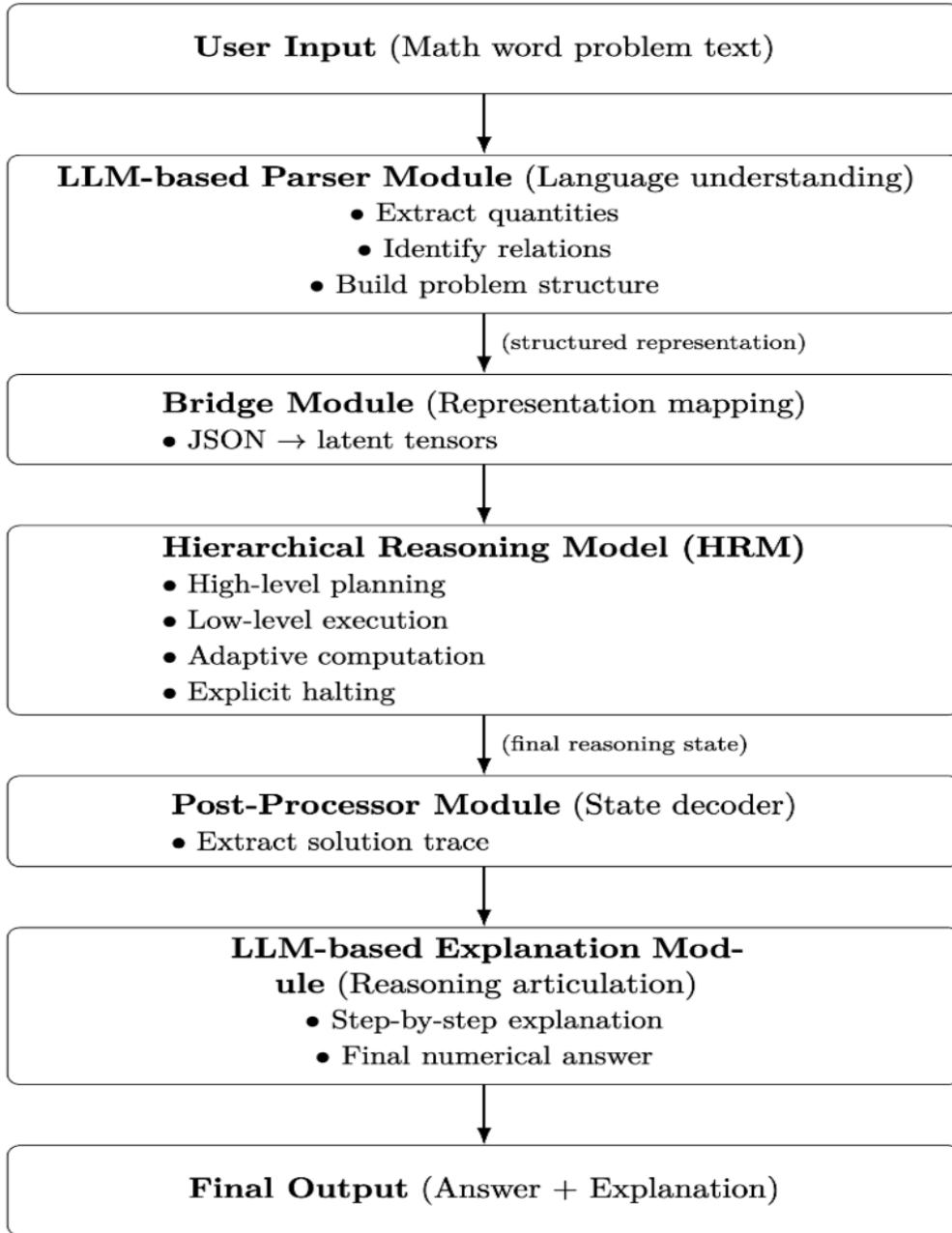


Figure 5.1: Top level architecture.

5.3 Component Design

Component 1 - LLM-based Parser Module (Language Understanding)

This is the entry point of the pipeline after the user's problem text arrives. The parser sends the problem text to a large language model and asks it to extract three things: the quantities mentioned in the problem, the relationships between those quantities, and the overall structure of the problem. The output is a structured JSON object.

The reason an LLM is used here - rather than a simpler rule-based parser - is that math word problems are written in many different ways. An LLM can handle this variation without needing hand-crafted rules for every phrasing. The JSON output is cached after the first call so the LLM does not need to be called again for the same problem during training.

The output of this module - labelled "structured representation" in Figure 5.1 - is passed to the Bridge Module.

Component 2 - Bridge Module (Representation Mapping)

The Bridge Module converts the structured JSON from the parser into a set of latent tensors that the HRM reasoning core can process. JSON is a human-readable text format - the HRM core works with numeric vectors. The Bridge Module is the connection between these two representations.

It reads each field in the JSON (variable names, numeric values, step descriptions) and encodes them into a fixed-size vector using a small feedforward network. The output is a latent tensor that represents the problem in a form the reasoning module understands.

The reason this is a separate module - rather than combining it with the parser or the HRM - is that it allows the representation mapping to be changed independently. If a different encoder is needed in future (for example, to handle a different input format), only this module needs to be replaced.

Component 3 - Hierarchical Reasoning Model (HRM Core)

This is the central reasoning component of the system. It takes the latent tensor from the Bridge Module and works through the problem in a two-level loop. [1]

The H-module (slow planner) runs once per outer cycle. It looks at the current state of the reasoning and decides what the next high-level step should be for example, whether to add, subtract, or compare quantities. The L-module (fast executor) then runs several steps under that plan, refining the internal state of the reasoning at a finer level. Adaptive Computation Time (ACT) allows the L-module to stop early on simple problems and run longer on harder ones.

This two-level design separates high-level planning from low-level computation. The reason this matters is that mathematical problems often require both deciding what operation to do next (planning) and then carrying it out precisely (execution). Keeping these two tasks in separate modules makes the reasoning more structured and easier to analyse.

After all outer cycles are complete, the HRM produces a final latent state labelled "final reasoning state" in Figure 5.1 which is passed to the Post-Processor Module.

Component 4 - Post-Processor Module (State Decoder)

The Post-Processor reads the final latent state from the HRM and extracts two things: the numeric answer to the problem, and the reasoning trace the sequence of intermediate states that the HRM passed through on its way to the answer.

The numeric answer is produced by a small decoder network that maps the final latent vector to a scalar value. The reasoning trace is the record of what the HRM computed at each step it is not generated separately but collected directly from the forward pass of the HRM.

The reason the trace is extracted here rather than inside the HRM is to keep the HRM core clean and focused on reasoning only. Trace extraction is a post-processing concern it does not affect the reasoning computation itself.

Component 5 - LLM-based Explanation Module (Reasoning Articulation)

The Explanation Module takes the reasoning trace from the Post-Processor and generates a step-by-step natural language explanation of how the answer was reached. It sends the trace to the large language model with a prompt asking it to write an explanation in plain English.

This module produces two outputs: the step-by-step explanation and the final numerical answer (formatted for display). Together these form the Final Output of the system.

This module is optional it can be switched off during training to save time and cost, and switched on during evaluation or when a human-readable explanation is needed. This design keeps the training pipeline lightweight while still supporting full explainability when required.

Component 6 — Final Output

The final output of the system is a pair: the numeric answer to the problem and a natural language explanation of how that answer was reached. This is the point where the system fulfils the interpretability goal a user can read both the answer and the explanation and decide whether the reasoning makes sense.

5.4 Design Rationale

Each design decision in this architecture was made to directly address part of the research gap identified in Chapter 2.

The structured JSON input addresses the problem of opaque embeddings. By converting the problem into an explicit representation of quantities and relations before it enters the reasoning module, the inputs become auditable a human can check what the model was given before reasoning started.

The separation of the parser, Bridge Module, and HRM into three distinct components means that each part can be tested and evaluated independently. If the system gives a wrong answer, it is possible to check the JSON output of the parser, then the tensor output of the Bridge Module,

then the HRM reasoning trace, to find exactly where the error occurred.

The LLM-based Explanation Module addresses the output interpretability gap. Instead of producing only a number, the system produces a readable explanation that a non-expert can follow.

5.5 Evaluation Design

The system will be evaluated on the GSM8K benchmark dataset , which contains 7,473 training problems and 1,319 test problems. The primary evaluation metric is exact-match accuracy the percentage of test problems where the system's predicted answer, rounded to the nearest integer, matches the correct answer exactly. [5]

Secondary metrics include:

- Mean Absolute Error (MAE) - the average difference between predicted and correct answers
- Median Absolute Error (MedAE) - the median difference, which is less sensitive to large errors on very hard problems
- Explanation coherence - a qualitative check that the generated explanation matches the steps the HRM actually took

5.6 Summary

This chapter described the full system architecture of the proposed hybrid pipeline. Each of the six components was explained in terms of what it does, why it was designed that way, and how it connects to the next component. The design keeps reasoning explicit and traceable at every stage from structured JSON input through HRM latent reasoning to natural language explanation output. Chapter 6 documents the implementation progress made so far.

6. Implementation

6.1 Introduction

This chapter explains how the proposed hybrid pipeline was built. Each module is described in terms of the software tools used, the hardware it runs on, the algorithm it follows, and how it connects to the rest of the system. Brief code snippets and pseudocode are included to show the key logic.

The entire system is implemented in a Kaggle notebook environment using Python 3.12 and PyTorch 2.11. All modules are organised as separate notebook cells that run in sequence from data loading through to evaluation.

6.2 Module 1 - Data Processing

Software: HuggingFace datasets library, Python regex, PyTorch DataLoader

Hardware: CPU preprocessing, Kaggle T4 GPU for tensor operations

What it does:

This module loads the GSM8K dataset and converts each problem into a structured format the model can use. The raw dataset contains a natural language problem and a chain-of-thought answer string ending with the final answer after a "#####" marker.

Algorithm:

1. Load the GSM8K train and test splits from HuggingFace [5]
2. Extract the final numeric answer using the "#####" marker
3. Parse the reasoning steps from the chain-of-thought lines
4. Build a structured JSON object for each problem containing: problem text, variable values, step descriptions, and the final answer
5. Apply log1p normalisation to all target answers for stable training

answer extraction

```
import re
def extract_answer(answer_text):
    match = re.search(r'####s*([\d,.-]+)', answer_text)
    return float(match.group(1).replace(',', '')) if match else None
```

Figure 6.1: Answer extraction from GSM8K solution strings using the #### delimiter pattern.

data pipeline (Pseudocode)

```
FOR each problem in GSM8K:
    answer = extract from "####" marker
    steps = parse CoT lines above "####"
    json = build {problem, variables, steps, answer}
    target = log1p(answer)
    store (json, target) in dataset
```

Figure 6.2: Pseudocode for GSM8K preprocessing.

6.3 Module 2 - LLM Parser and Cache

Software: NVIDIA NIM API (OpenAI-compatible), Python json library, file-based cache

Hardware: External API call (no local GPU required), cache stored on Kaggle disk

What it does:

This module sends each GSM8K problem to a large language model and asks it to return a structured JSON object containing the quantities, variable names, and step structure of the problem. The result is saved to a local cache file so the API is only called once per problem.

Algorithm:

1. For each problem, check if a cached JSON file already exists
2. If yes, load from cache and return
3. If no, send the problem text to the LLM API with a prompt requesting JSON-only output

4. Clean and validate the returned JSON
5. If parsing fails, retry up to 3 times; fall back to deterministic parser if all retries fail
6. Save the result to the cache file

Code snippet (LLM call with retry):

```
def parse_with_llm(problem_text, retries=3):
    for attempt in range(retries):
        try:
            response = client.chat.completions.create(
                model="nvidia/llama-3.1-70b-instruct",
                messages=[{"role": "user",
                           "content": f"Return JSON only:\n{problem_text}"}]
            )
            return json.loads(response.choices[0].message.content)
        except Exception:
            time.sleep(2)
    return fallback_parser(problem_text)
```

Figure 6.3: LLM-based problem parser using NVIDIA NIM.

6.4 Module 3 - JSON Encoder (Bridge Module)

Software: PyTorch nn.Module, custom feedforward network

Hardware: Kaggle T4 GPU

What it does:

This module reads the structured JSON produced by the parser and converts it into a fixed-size latent tensor that the HRM core can process. It encodes numeric variable values and operation type labels into separate embeddings, then combines them into one vector.

Algorithm:

1. Read numeric variable values from the JSON

2. Pass each value through a small feedforward network to produce a value embedding
3. Look up the operation type (add, subtract, multiply, etc.) in an embedding table
4. Concatenate value and operation embeddings
5. Pass through a linear projection layer to produce the final latent tensor

Pseudocode

```

FOR each variable in JSON["variables"]:
    val_emb = Linear(variable.value)
    op_emb = EmbeddingTable(variable.operation)
    step_emb = concat(val_emb, op_emb)
latent = Linear(mean_pool(all step_embs))
RETURN latent tensor shape (B, seq_len, hidden_dim)

```

Figure 6.4 : encoding structured JSON variables into a latent tensor

6.5 Module 4 - HRM Core (Reasoning Module)

Software: PyTorch nn.Module, multi-head attention, RMSNorm, SwiGLU activation

Hardware: Kaggle T4 GPU (16 GB VRAM)

What it does:

This is the central reasoning component. It takes the latent tensor from the Bridge Module and runs it through a two-level loop , an outer H-loop for high-level planning and an inner L-loop for step-by-step execution until either all cycles are complete or the halt gate triggers early exit.

Algorithm

```

z = latent tensor from Bridge Module
FOR n = 1 to N (outer H-cycles):
    plan = H_module(z)      # slow planner
    state = z + project(plan) # inject plan into state

```

```

halt_accum = 0
FOR t = 1 to T (inner L-steps):
    state = L_module(state) # fast executor
    halt_prob = sigmoid(halt_gate(state))
    halt_accum += halt_prob
    IF halt_accum / t > threshold:
        BREAK # ACT early exit
z = detach(state) # one-step gradient approx
aux_pred[n] = aux_decoder(z) # deep supervision
RETURN z (final reasoning state), aux_preds

```

Figure 6.5: nested H-module (planner) and L-module (executor) loop with ACT early exit, one-step gradient detachment, and per-H-step auxiliary supervision.

Code snippet (ACT halt gate):

```

halt_accum = 0.0
for t in range(self.T):
    state = self.l_module(state)
    halt_prob = torch.sigmoid(self.halt_gate(state)).mean()
    halt_accum += halt_prob
    if (halt_accum / (t + 1)).item() > self.halt_threshold:
        break

```

Figure 6.6 : ACT early exit in the L-module .

6.6 Module 5 - Decoder and Post-Processor

Software: PyTorch nn.Module, attention pooling, linear layers

Hardware: Kaggle T4 GPU

What it does:

The decoder reads the final latent state from the HRM core and produces a numeric answer prediction and a confidence score. The post-processor extracts the reasoning trace the sequence of intermediate states and formats it for the explanation module.

Algorithm:

1. Apply attention pooling over the final latent sequence to get a single vector
2. Pass through a 3-layer MLP to predict the normalised answer
3. Apply log1p denormalisation to convert back to the original answer scale
4. Pass through a separate confidence head to produce a score between 0 and 1
5. Collect the aux_pred values from each H-step as the reasoning trace

Code snippet (answer decoding):

```
pooled = self.attention_pool(z_final)    # (B, hidden)
pred_norm = self.answer_head(pooled)    # (B, 1)
pred = torch.expml(pred_norm.clamp(-10, 10)) # denormalise
conf = torch.sigmoid(self.conf_head(pooled)) # (B, 1)
```

Figure 6.7: Decoder pseudocode .

6.7 Module 6 - Training Loop

Software: PyTorch training loop, Adam-atan2 optimiser, OneCycleLR scheduler, tqdm

Hardware: Kaggle T4 GPU, 40 epochs, batch size 32

What it does:

The training loop ties all modules together. It builds the DataLoader, runs forward passes, computes the combined loss, runs backpropagation, and saves checkpoints. The best model by validation accuracy is kept.

Code snippet (answer decoding):

```
Total Loss = SmoothL1(pred, target)
```



```
+ 0.1 * MSE(confidence, exp(-|pred - target|))  
+ 0.05 * mean(SmoothL1(aux_pred[n], target) for n in N)  
+ 0.01 * RL_loss (applied from epoch 10 onwards)
```

Figure 6.8 : Answer Decoding.

Pseudocode (training loop):

```
FOR epoch = 1 to 40:  
  FOR each batch in train_loader:  
    pred, conf, aux_preds = model(json_input)  
    loss = smooth_l1(pred, target)  
    + 0.1 * mse(conf, reward_signal)  
    + 0.05 * aux_loss(aux_preds, target)  
    loss.backward()  
    clip_grad_norm(model, max_norm=1.0)  
    optimiser.step()  
    scheduler.step()  
  val_acc = evaluate(val_loader)  
  IF val_acc > best_acc:  
    save_checkpoint(model)
```

Figure 6.9: Training loop pseudocode — composite loss (SmoothL1 + 0.1×MSE confidence + 0.05×auxiliary), gradient clipping, OneCycleLR scheduling, and best-checkpoint saving over 40 epochs..

6.8 Module 7 — Evaluation and Explanation

Software: PyTorch inference, NVIDIA NIM API for explanation, Python

Hardware: Kaggle T4 GPU for inference; API for explanation generation

What it does:

After training, the best checkpoint is loaded and run on the full GSM8K test set. Predictions are denormalised and compared to the correct answers. The reasoning trace from each problem is optionally sent to the LLM explanation module to generate a step-by-step natural language explanation.

Metrics computed:

- Exact-match accuracy: percentage of predictions where $\text{round}(\text{pred}) == \text{round}(\text{true answer})$
- Mean Absolute Error (MAE)
- Median Absolute Error (MedAE)

Current status:

The HRM core has been fully implemented and training is in progress.. Full evaluation on the GSM8K test set is pending completion of the 40-epoch training run and resolution of the representation collapse issue identified.

6.9 Implementation Progress Summary

Module	Status
Data Processing (GSM8K loading, JSON building)	Complete
LLM Parser and cache builder	Complete
JSON Encoder (Bridge Module)	Complete
HRM Core (H-module, L-module, ACT, aux decoders)	Complete
Decoder and Post-Processor	Complete
Training Loop (loss, optimiser, checkpointing)	Complete
Full 40-epoch training run	Pending
Evaluation on GSM8K test set	Pending
LLM Explanation Module integration	Pending

Figure 7.0: Implementation progress.

6.10 Evaluation Plan

The full evaluation of the hybrid pipeline will be conducted in the final thesis once the model reaches the desired accuracy on the validation set. The evaluation will follow the strategy below.

Dataset: GSM8K test split - 1,319 problems, held out during all training runs.

Primary metric: Exact-match accuracy - the percentage of test problems where the predicted answer, rounded to the nearest integer, matches the correct answer exactly. This is the standard benchmark metric for GSM8K. [5]

Secondary metrics: Mean Absolute Error (MAE), Median Absolute Error (MedAE), and validation loss, to measure how close predictions are even when they do not exactly match.

6.11 Remaining Implementation Plan

Week	Milestone	Description
Week 1	Resolve scale mismatch	Re-initialise decoder output layer with log1p-scale mean, fix expm1 clamping to prevent extreme predictions
Week 2–3	Complete full training run	Run the full training process on GSM8K, monitor performance each epoch and save the best-performing checkpoint.
Week 4–5	Integrate LLM parser module	Replace deterministic CoT parser with NVIDIA NIM LLM parser, rebuild JSON cache for full GSM8K train split
Week 5–6	Integrate explanation module	Connect LLM explanation module to trained HRM, generate and verify step-by-step explanations on test samples
Week 6–7	Full pipeline evaluation	Run complete hybrid pipeline on GSM8K test split, compute exact-match accuracy, MAE, MedAE, and explanation coherence
Week 8–10	Thesis writing	Write evaluation chapter with results and baseline comparisons, write conclusion chapter, submit final thesis

Figure 7.1: Further work

6.11 Summary

This chapter described how each module of the hybrid pipeline was designed and built. All six core modules - the data processor, LLM parser and cache, JSON encoder, HRM reasoning core, decoder, and training loop have been fully implemented. The training pipeline is running on the GSM8K dataset and producing decreasing loss values across epochs, confirming that the model is learning. A scale mismatch between the decoder output and the true answer range has been identified and a targeted fix is planned as the first priority in Week 1. The evaluation plan in Section 6.10 describes how the complete system will be tested against baselines in the final thesis. The remaining work outlined in Section 6.11 is well-defined and time-bounded, keeping the project on track for final thesis submission.

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